

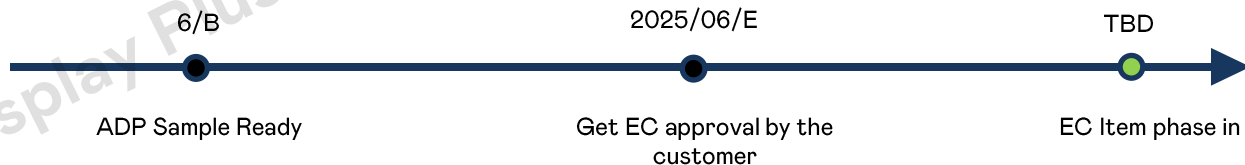


G320ZAN02.3 Engineering Change Report

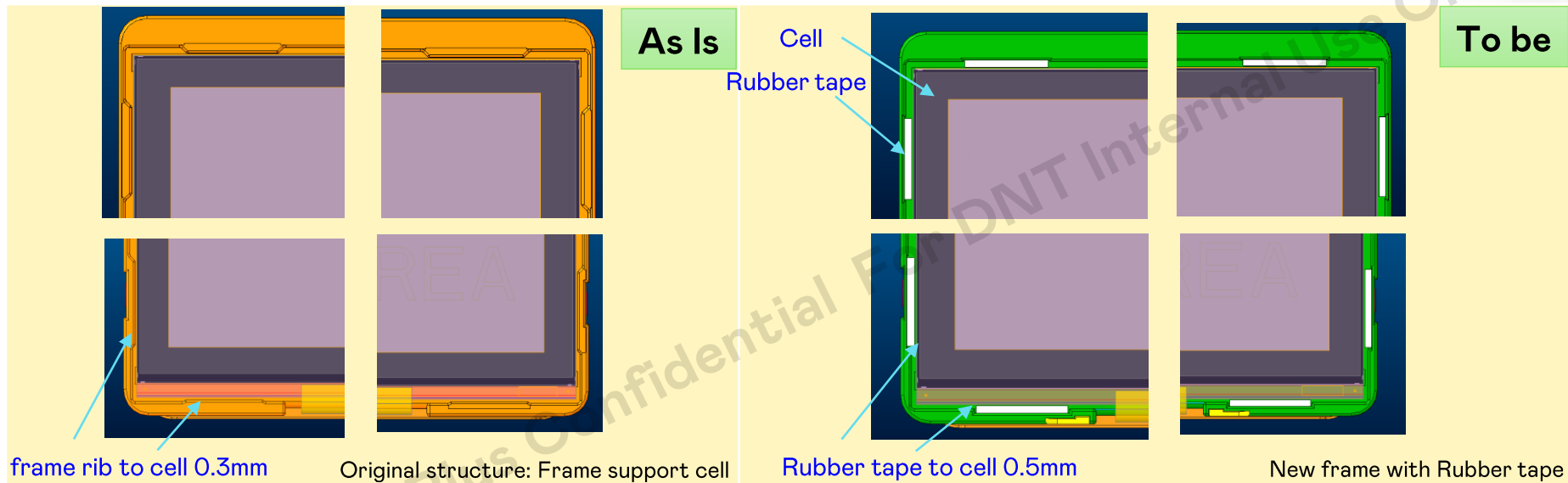
GDIS

G320ZAN02.3 EC summary

EC ITEM	Sample send out	Get EC approval by the customer	EC Item phase in
Modify the frame structure	2025/6/B	2025/6/E	TBD (ADP will notify DNT in advance before switching.)
Cell tape cross-posting			
Add front bezel punch			
Red Line			
UV328 change to UV928			



EC item : New BLU plastic frame

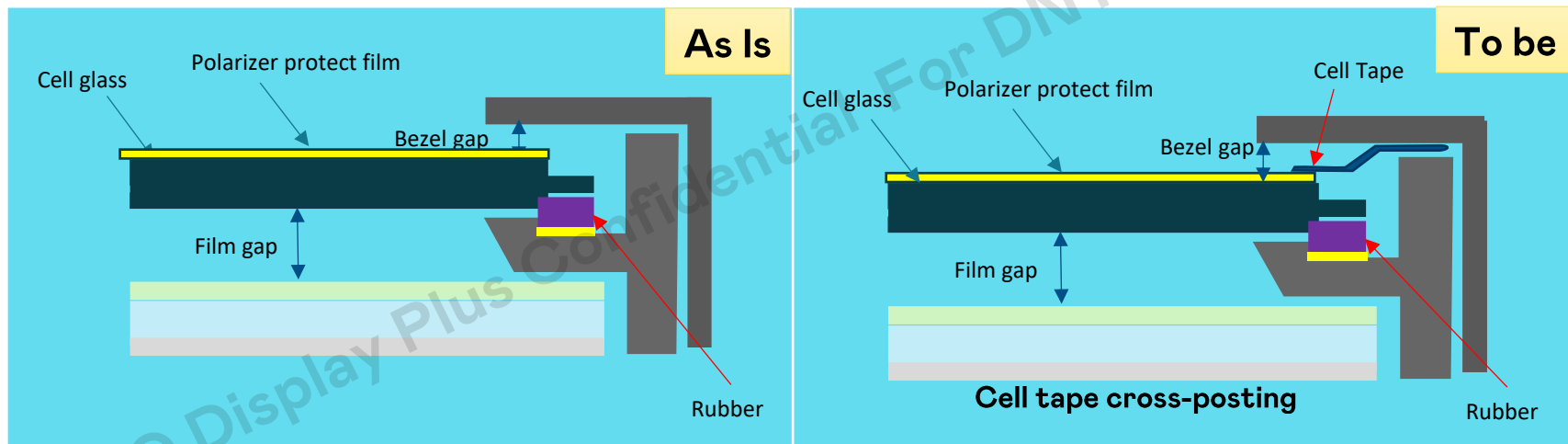


- The continuous scratching between the raised part of the plastic frame on the BLU and the cell, which cause the tiny scrap interference to the BLU module.

- Due to the Rubber tape on the new BLU plastic frame, the Rubber tape smooth contact surfaces with new BLU plastic frame after continuous scratching won't produce the tiny scrap.

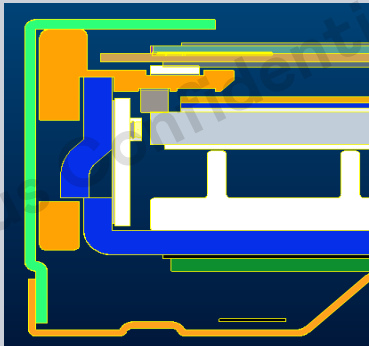
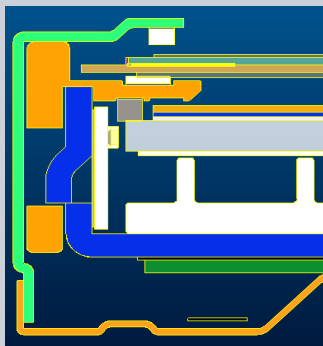
ECR item : Cell tape cross-posting

- Background: Because of 32" Series without Dustproof design, Foreign matter is always happened between Cell and BLU Top film. It may cause High failure ratio and More Cleaning time in production line.
- Root Cause : Because 32" are disassembled for Cover lens lamination, Dustproof design_Cell fixed on BLU Frame by double side tapes at that time can't be implemented/accepted by customer.
- Material structure comparison



ECR item : Add front bezel punch

- **Action:** Add front bezel punch let customers could cover it back after DB process .

	As Is	To be
From bezel		

ECR item : Red line

■ Root cause

1. Figure 1 shows the array design layout. Due to process deviation, GOA pad and red subpixel is close to each other.
2. As below figure 2, the cross section of GOA pad and subpixel, when panel power-on, the GOA pad induces charge on red pixel electrode edge.
3. The induced charge causes the extra electric field to make liquid crystal twist a little bit different.
4. After power off, the induced charge released by time so that red line disappear.
5. As we input white frame pattern(right side), nothing abnormal. It means function OK.

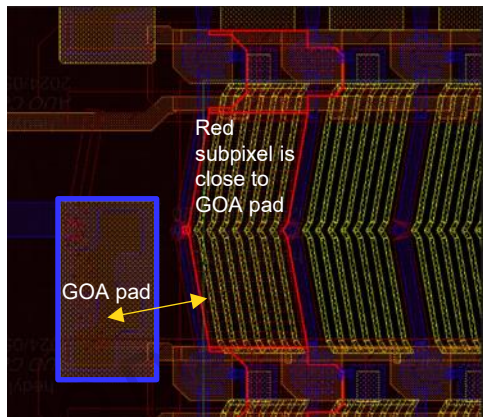
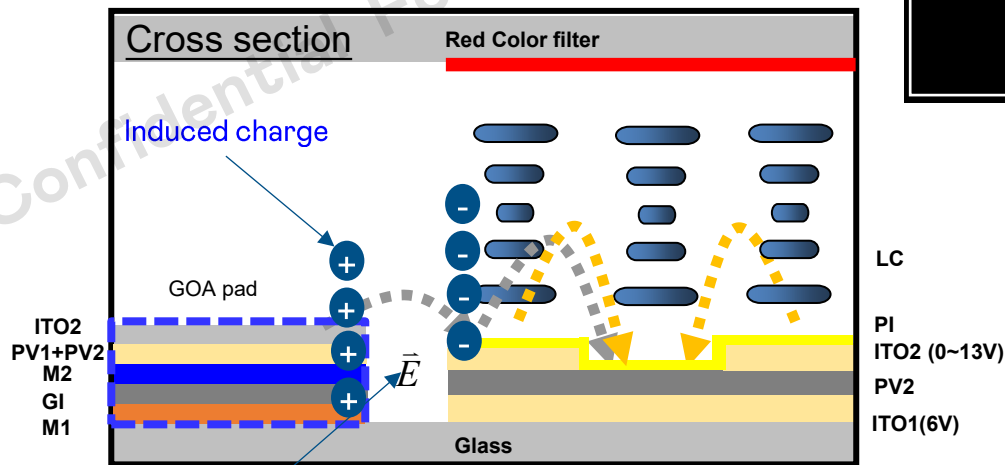


Figure 1



extra electric field

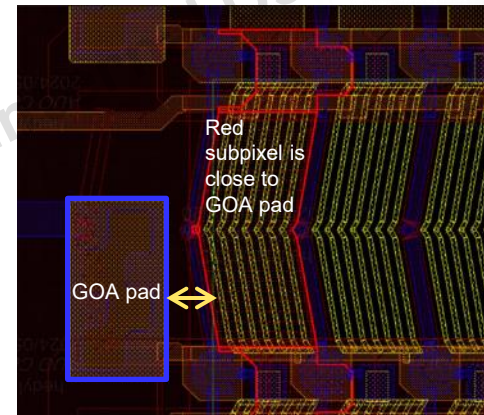
Figure 2



ECR item : Red line

❑ Countermeasure

1. Distance between GOA pad and red subpixel is $25 \pm 5 \mu\text{m}$.
2. Enlarge the distance by process control (photo exposure, etching, layer thickness.....)



ECR item: UV 328

- **Cause:** May 1st-12th, 2023, the Conference of the Parties to the Stockholm Convention held its 11th meeting and agreed to list UV-328 (CAS 25973-55-1) under Annex A for global elimination. For display, UV-328 is generally applied to Tri-acetyl Cellulose (TAC) film, one of layers of polarizer. Fortunately, the TAC film belongs to the category of 5-year exemption and AUO Display Plus (ADP) can keep supplying displays containing UV-328.
- **Action:** To make the engineering change proceed efficiently, the running mode will be different from usual. The UV-328 free polarizers themselves will be qualified by checking their optical properties and reliability on few display models. After qualification, UV-328 free polarizers will be adopted to all impacted models. Because of the enormous number of impacted models, we are not able to provide engineering samples and expect customers could approve this engineering change based on the validation results that ADP provides.

EC Import time and switching method

Desired Implementation Method & Date

By new part number to distinguish/
97.32G02.300→97D32G02.302



Tap Into The Possibilities